Tanya Singh

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EDUCATION

University of Maryland, College Park **College Park, MD** **Graduation Date: May 2027**

Bachelor of Science in Computer Science, Minor in General Business *GPA: 3.985, Dean's List*

- Relevant Coursework: Object-Oriented Programming, Applied Probability and Statistics, Discrete Structures, Computer Systems, Linear Algebra, Organization of Programming Languages, Algorithms, Data Science, Advanced Data Structures, Machine Learning, Compilers
- Leadership: Committee Member at Association for Women in Computing, Marketing Lead at Google Developer Student Club, Committee Head at Campus Coders Crew, Researcher in the First-Year Innovation & Research Experience

SKILLS

- **Languages:** Python, Java, C, SQL, R, OCaml, Rust, JavaScript, HTML, CSS, MATLAB, x86 Assembly, Haskell, Racket, Dafny
- **Libraries:** Scikit-learn, Pandas, NumPy, SciPy, Matplotlib, Seaborn
- **Tools & Platforms:** Microsoft 365 Copilot Agent Builder, Git/GitHub, Azure DevOps, ServiceNow, Microsoft Excel, VS Code, RStudio
- **Concepts:** Machine Learning (Classification, Clustering, Regression), Large Language Models, Prompt Engineering, AI Agent Development, Knowledge Base Design, Hypothesis Testing, Data Visualization, Data Analysis, Agile/Scrum, Zero Trust
- **Certifications:** Microsoft Office Specialist (Associate)

WORK EXPERIENCE

Navy Federal Credit Union **Vienna, VA** **May 2026 - August 2026**

Summer Associate

- Deployed an M365 Copilot-powered Zero Trust AI Assistant that provides cited answers on 29 active projects, 45 capabilities, and 152 activities.
- Designed a metadata-driven knowledge base by cataloging 84 Zero Trust documents and standardizing AI retrieval across teams and workstreams.
- Leveraged Azure DevOps and ServiceNow data to connect Zero Trust activities with active projects, enabling visibility within the framework.
- Reduced AI response times by 90%+ by cutting retrieval latency from 2+ minutes to 10 seconds through prompt engineering and optimization.

Lavner Education **Washington, DC** **June 2025 - August 2025**

Intern/Instructor

- Taught 120+ students coding, robotics, and STEM concepts through hands-on projects, adapting instruction to different age groups and skill levels.
- Managed software installations, classroom computer systems, and student check-in operations, contributing to a reported 20% improvement in operational efficiency.

Easy Dynamics **McLean, VA** **January 2025 - January 2025**

Intern

- Evaluated 4+ LLMs for researching payroll fraud and security threats to asses the ability to retrieve and summarize publicly available information.
- Developed 12+ research prompts to compare model accuracy, quality, and efficiency, documenting differences in performance across models.
- Presented evaluation results and recommendations for integrating LLMs into security research workflows using Microsoft applications.

Code Ninjas **Leesburg, VA** **November 2022 - November 2023**

Code Sensei

- Taught 50+ students ages 5-14 programming fundamentals in Scratch, JavaScript, and C#, guiding them through coding exercises and debugging.
- Mentored students on game development projects using 5+ platforms, including Unity, Roblox Studio, and MCreator.

PROJECTS

Machine Learning Stroke Risk Prediction | *Python* **2026**

- Developed an end-to-end stroke prediction pipeline using Python, Pandas, NumPy, and Scikit-learn, preprocessing 5,110 patient records through data cleaning, feature engineering, and scaling. Trained and compared Random Forest and Logistic Regression models to examine the influence of age, glucose levels, and BMI on stroke risk.

Lexer, Parser, and Interpreter | *OCaml* **2025**

- Built a custom programming language implementation in OCaml, developing a lexer for tokenization, a recursive-descent parser for syntax analysis, and an interpreter for program evaluation. Implemented lexical scoping, closures, and environment-based evaluation to support variable bindings and function execution.

Performance Optimization & Multithreading | *C* **2025**

- Developed a custom dynamic memory allocator in C to implement explicit memory allocation and management. Built a parallel matrix normalizer using pthreads, mutexes, and barriers to distribute computations across multiple threads, coordinate shared memory access, and synchronize concurrent operations.

Personal Portfolio | *HTML, CSS, JavaScript* **2025**

- Designed, developed, and deployed a responsive portfolio website using HTML, CSS, and JavaScript, hosted on a custom domain. Implemented interactive animations, dynamic project filtering, and dedicated pages for technical projects, professional experience, research, and coursework.

RESEARCH

Effectiveness of Large Language Models (LLMs) in Weather Forecasting for Arlington, VA | *R* **2025**

- Co-authored a research study evaluating GPT-5, Claude Sonnet 4, and Gemini 2.5 Pro on 92 days of Arlington, VA weather forecasts using one year of historical data. Processed model outputs, conducted paired t-tests, and performed Wilcoxon signed-rank tests in R, identifying statistically significant differences in high-temperature predictions ($p < 0.001$), while GPT-5's low-temperature predictions showed no statistically significant difference from observed values ($p = 0.66$).